

Multi-Source Pre-Match Signal Fusion and Uncertainty-Driven Risk Control for Football Outcome Forecasting: A Leakage-Controlled Evaluation of Machine Learning and Large Language Models

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Abstract

In sports betting, the closing odds of major bookmakers are widely regarded as an efficient aggregation of public information, making predictive gains over the market difficult to achieve and even harder to verify. We present a leakage-controlled evaluation of statistical, machine-learning, and large-language-model (LLM) methods for football match outcome forecasting, built on 11,811 matches from the five major European leagues across seven seasons (2019/20–2025/26), with a held-out test season of 1,104 matches. All features are restricted to pre-match information, all normalization statistics are fit on the training split only, and every reported quantity is accompanied by a bootstrap confidence interval. *The test season (2025/26) is incomplete (through 2026-02-12): financial point estimates may shift when the season completes, so ROI values should be read as indicative rather than definitive.* Three findings stand out. First, neither tree-based models nor a domain-enhanced LLM exceed the predictive accuracy of de-vigged closing odds (54.2% accuracy, 0.967 log loss): the market remains unbeaten, and the LLM matches market-level probability calibration at a cost of \$0.55 for the entire test season. Second, permutation importance and ablations show that market signals carry roughly 87% of the predictive information, while referee statistics provide no measurable gain. Third, and most importantly, an uncertainty index built from prediction confidence and market volatility reliably stratifies risk: low-uncertainty matches are

predicted at 68–69% accuracy, whereas matches flagged as high uncertainty lose 9.6–13% ROI and are best skipped by a no-bet policy. Uncertainty filtering improves risk-adjusted returns, whereas value-based selection of high-confidence bets systematically loses money, indicating that apparent model edge is overconfidence rather than information. These results provide a rigorous benchmark for sports forecasting research and evidence that uncertainty-driven risk control, not accuracy, is the lever that connects prediction quality to financial stability.

1 Introduction

Sports outcome forecasting has attracted sustained attention from statistics, machine learning, and, more recently, large language models (LLMs). The task is attractive because outcomes are public, structured, and frequent, and because a predictive model can be evaluated not only on accuracy but also on downstream financial consequences in betting markets. However, the same properties make football forecasting a demanding test bed: betting markets aggregate vast amounts of information into closing odds, and any method that hopes to add value must do so *against* a strong, continuously updated baseline.

The literature has moved in two directions. On the modeling side, classical approaches—Poisson and Dixon–Coles models, Elo ratings, and gradient-boosted trees—provide strong and interpretable baselines. On the representation side, LLM-based forecasting sys-

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tems [1, 2, 3] demonstrate that language models can produce calibrated probability forecasts when given structured evidence. Yet most published evaluations in the LLM forecasting line measure accuracy and Brier scores in isolation, without a market baseline, without explicit leakage controls, and without a betting protocol that would make the financial claims reproducible.

In this paper we address these gaps with a deliberately conservative evaluation design. Our contributions are as follows.

- **A leakage-controlled benchmark against the market.** We assemble 11,811 matches from the top five European leagues (2019/20–2025/26), restrict all features to information available before kickoff, fit every statistic on the training split only, and evaluate on a held-out future season (1,104 matches). Against this backdrop we compare de-vigged closing odds, Poisson and Dixon–Coles models, Elo, random forests, gradient boosting, and a domain-enhanced LLM. None of the learned methods exceeds market accuracy; the LLM matches market-level probability calibration at negligible cost.
- **An interpretable decomposition of predictive information.** Permutation importance and ablations show that market signals account for approximately 87% of predictive information, expected-goals (xG) features add a small but consistent gain, and referee statistics provide no measurable improvement.
- **Empirical validation of uncertainty-driven risk control.** We define a corrected scenario-complexity score and an uncertainty index, and show on held-out data that they stratify matches monotonically: low-uncertainty matches reach 68–69% accuracy while high-uncertainty matches lose 9.6–13% ROI. A no-bet policy on high-uncertainty matches improves risk-adjusted returns, whereas value-based selection of high-confidence bets loses money—evidence that apparent model edge reflects overconfidence, not information.
- **A reproducible betting protocol.** All financial evaluations use equal-stake or fractional-Kelly sizing on a single bookmaker’s closing odds, with bootstrap confidence intervals, coverage reporting, and no-bet decisions separated from executed bets.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the data, features, models, and risk-control framework. Section 4 presents the experimental results. Section 5 discusses implications and limitations, and Section 6 concludes.

2 Related Work

2.1 Statistical and Machine-Learning Models for Football

The canonical generative model for football scores is the Poisson model of Maher [4], refined by Dixon and Coles [5], who introduce a low-score correction and time-decayed team parameters. Elo-style ratings [6] remain a strong and transparent baseline, and gradient-boosted decision trees have become the standard tabular approach in sports analytics. A consistent finding of this literature is that predictive accuracy saturates around 50–55% for three-way match outcomes, close to the empirical base rates, and that bookmaker odds are difficult to outperform [7, 8].

2.2 Market Efficiency and Closing Odds

Bookmaker closing odds are widely regarded as near-efficient aggregators of public information [9, 10]. In particular, the closing line is a stronger predictor than the opening line, and drift from opening to closing odds contains little exploitable information once transaction costs are considered. Any claimed predictive edge must therefore be evaluated against de-vigged closing odds under an identical protocol; evaluations that omit this baseline cannot distinguish model skill from market inefficiency. Our work follows this prescription, and additionally reports the opening-to-closing drift experiment as a direct test of market efficiency on our data.

2.3 LLM Probabilistic Forecasting

Large language models have recently been shown to produce calibrated probabilistic forecasts from textual evidence. Halawi et al. [1] demonstrate that fine-tuned LLMs approach human expert forecasting on general prediction markets, and the AIA Forecaster [2] scales this to a multi-agent retrieval-augmented system. LLM-as-a-Prophet [3] provides a controlled “prophet arena” in which the predictive intelligence of LLMs can be benchmarked. These systems rely on web retrieval and fine-tuning, which raises the bar for reproducibility and makes leakage control difficult to audit. Our setting is deliberately simpler: the LLM receives only a pre-match feature card, has no retrieval access, and is evaluated against the market under the same protocol as all other models. This isolates the contribution of domain-enhanced reasoning from retrieval and training data effects.

2.4 Gaps

Three gaps motivate our design. (i) Most evaluations lack a de-vigged market baseline under an iden-

tical betting protocol. (ii) Leakage controls—pre-match information cutoffs, train-only statistics, temporal splits—are rarely stated precisely enough to reproduce. (iii) Uncertainty and risk are either absent or defined by hand-crafted scores whose empirical validity is not tested. We address all three with a conservative, fully specified evaluation and with empirically validated uncertainty signals.

3 Methodology

3.1 Data

Figure 1 gives an overview of the full pipeline; the components are detailed below. We use 11,811 league matches from the English Premier League (E0), Spanish La Liga (SP1), German Bundesliga (D1), Italian Serie A (I1), and French Ligue 1 (F1), covering the 2019/20–2025/26 seasons (2025/26 is incomplete, ending 2026-02-12; we report this honestly). Each match record contains results, referee, and opening and closing odds from eight bookmakers (Bet365, Bet&Win, Interwetten, Pinnacle, William Hill, Ladbrokes, Max, and average) in football-data.co.uk format. We additionally crawl per-match expected goals (xG, xGA, npG, npGA), pressing intensity (PPDA), and deep progression from Understat for all five leagues and all seven seasons (24,918 team-match rows; 99.3%/99.2% of home/away sides match league-match dates after name normalization).

Leakage control. The evaluation is strictly chronological: train = 2019/20–2023/24 (8,955 matches), validation = 2024/25 (1,752), test = 2025/26 (1,104). All features are constructed from information available before kickoff; rolling statistics use a shift of one to exclude the current match; season standings are computed before kickoff; scaler statistics and missing-value medians are fit on the training split only and applied to validation and test. A manual verification script recomputes a sample of features from raw records to confirm the absence of leakage.

3.2 Features

All features are pre-match:

- **Team form:** rolling (5-match) means of goals for/against, points, win rate, shots, shots on target, corners, fouls; season cumulative points, goal difference, games, points per game, and league rank before kickoff (home and away separately).
- **xG time series:** rolling (5-match) means of xG, xGA, npG, npGA, PPDA coefficient, deep progressions, and expected points, plus the home-away xG difference.

- **Market signals:** de-vigged implied probabilities from closing odds (Bet365 and average), opening-to-closing odds movement for home/draw/away, cross-bookmaker dispersion of closing odds (σ_{market}), and opening average odds.
- **Referee context:** the referee’s historical pre-match averages of cards, red cards, and fouls (expanding window over the referee’s prior matches; available for the Premier League only).

This yields 60 features in total.

3.3 Models

- **Market baseline:** de-vigged closing odds (probabilities obtained by inverting odds and renormalizing). This is the central baseline under the same evaluation protocol.
- **Poisson** and **Dixon–Coles** models fit per league by maximum likelihood on the training split.
- **Elo** ratings with home advantage, mapped to probabilities by multinomial logistic regression fit on training.
- **Random Forest** and **XGBoost** trained on the 60 pre-match features with early stopping on the validation split.
- **Domain-enhanced LLM:** DeepSeek-Chat, prompted with a pre-match feature card (team form, rank, market signals, referee context), a system prompt stating that closing odds are a strong baseline and forbidding invented information, and a chain-of-thought request that outputs a calibrated probability triple in JSON. The LLM has no retrieval access. Three samples are drawn per match at temperature 0.3; the mean is the prediction and the pairwise total-variation agreement is the consistency score.

3.4 Scenario Complexity Score

The scenario complexity score measures how hard a match is to predict from *market information alone*. Unlike earlier formulations in which a larger strength gap was treated as *more* complex (a direction error), we define

$$\text{SCS} = (1 - \text{gap}_{\text{norm}}) + \text{vol}_{\text{norm}} + \text{move}_{\text{norm}},$$

where gap_{norm} is the normalized absolute league-rank difference between home and away teams (larger gap $\Rightarrow$ easier to predict), vol_{norm} is the robust-normalized cross-bookmaker dispersion of closing odds, and $\text{move}_{\text{norm}}$ is the robust-normalized opening-to-closing odds movement (a proxy for late information

arrival). All components lie in $[0, 1]$ and are fit on the training split. A larger SCS indicates a more uncertain match.

3.5 Uncertainty Index and Risk Control

The uncertainty index is conceptually distinct from the SCS: SCS is a *purely market-side* measure of scenario complexity (rank gap, odds dispersion, odds drift), whereas the UI *fuses model output with market signals*—predictive confidence and multi-sample consistency from the model, plus the same normalized market-volatility and odds-movement terms. The shared market terms are intentional: the UI inherits market uncertainty and adds the model’s own confidence signal. The index combines these components as

$$UI = w_1(1 - p_{\max}) + w_2(1 - C) + w_3 \text{vol}_{\text{norm}} + w_4 \text{move}_{\text{norm}},$$

with default weights $(0.4, 0.3, 0.15, 0.15)$; $p_{\max}$ is the model’s maximum predicted probability and C the multi-sample consistency (1 for deterministic models). Risk tiers are $UI \leq t_{\text{low}}$ (full stake), $t_{\text{low}} < UI \leq t_{\text{high}}$ (linearly scaled stake), and $UI > t_{\text{high}}$ (no-bet). Thresholds are selected on the validation split ($t_{\text{low}} = 0.30$, $t_{\text{high}} = 0.45$ given the observed UI distribution) and reported with a sensitivity analysis. A stop-loss rule reduces the stake to 20% after five consecutive losses or a 10% drawdown from peak bankroll.

3.6 Betting Protocol and Metrics

All financial simulations are fully specified: equal stakes of one unit (or fractional Kelly capped at 10% of bankroll where stated), the model’s maximum-probability outcome, Bet365 closing odds for settlement, sequential bankroll updating, and no-bet decisions excluded from executed-bet statistics but reported separately with coverage. Predictive metrics are accuracy, macro-F1, log loss, Brier score, and expected calibration error (ECE). Financial metrics are ROI, Sharpe ratio, maximum drawdown, and win rate. All main quantities are reported with bootstrap 95% confidence intervals.

4 Experiments and Results

4.1 Overall Performance

Table 1 reports the overall performance on the held-out test season. Several observations are immediate. First, the de-vigged closing odds achieve 54.2% accuracy and a log loss of 0.967, and no learned model exceeds them in accuracy within statistical uncertainty (bootstrap 95% confidence intervals for accuracy all overlap the

market point estimate; the largest gap, Random Forest at 54.7%, is well inside the market’s CI of $[51.4\%, 57.2\%]$). Second, the domain-enhanced LLM reaches a log loss of 0.967 and ECE of 0.025, statistically indistinguishable from the market (0.967 and 0.026): a zero-training prompt reaches market-level probability quality. The reliability diagram in Figure 2 confirms this: all three probability sources track the diagonal, with the LLM and the market showing the smallest deviation. Third, all ROI values are small and their confidence intervals cross zero: no method, including the market itself, shows a stable profit over this test season under equal-stake betting. This is the expected regime of an efficient market, and it is the backdrop against which all subsequent analyses are interpreted. We remind the reader that the 2025/26 test season is incomplete (matches through 2026-02-12): accuracy estimates are stable (they depend on many games), but the financial point estimates above may shift when the season completes, and the bootstrap intervals should be read accordingly.

4.2 Ablation Study

Table 2 ablates feature groups from XGBoost. Removing market odds costs 3.1 percentage points of accuracy (54.2% to 51.1%) and 0.028 of log loss, confirming that bookmaker information dominates. Removing referee statistics changes nothing (54.2% vs 54.2%): the referee features provide no measurable predictive gain, a negative result we report deliberately. (One caveat: referee records exist only for the Premier League in our data, so the referee feature group is evaluated on a subset of matches; the null result should be read as *no gain where referee data exist*, not as a global statement about referee information.) xG features contribute a small but consistent improvement (removing them raises log loss slightly), and rank and recent-form features contribute little beyond what the market already encodes.

4.3 Feature Importance

Permutation importance (Table 3) quantifies the same hierarchy: market-derived features account for roughly 87% of the total predictive information, xG for about 6%, and referee, rank, and form features for under 2% combined. The result is robust across the ablation and permutation analyses and motivates the position of this paper: the marginal information left to any forecasting model is small and concentrated in a few market-driven signals.

4.4 League and Season Generalization

Performance varies markedly across leagues (Table 4): the Bundesliga and La Liga are the most predictable

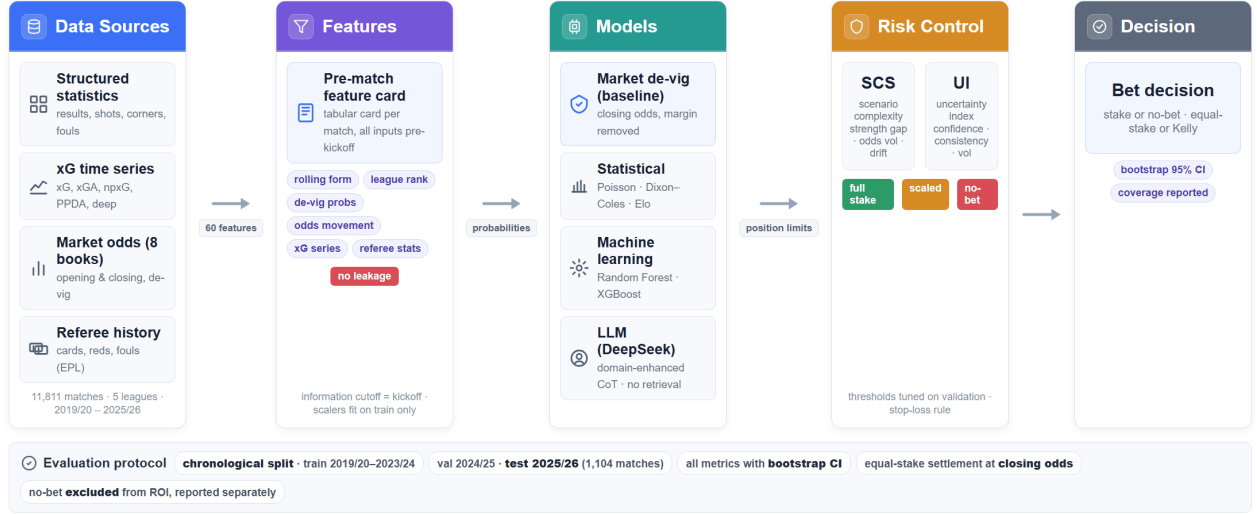


Figure 1: Overview of the proposed framework. All features are constructed from pre-match information only; every model is evaluated against de-vigged closing odds under an identical protocol; and the risk-control layer converts model uncertainty into stake sizing and no-bet decisions.

Table 1: Overall performance on the held-out test season (1,104 matches). The LLM parsed 100% of matches at a cost of \$0.55. Bootstrap 95% confidence intervals are reported in the text.

Model	Acc	LogLoss	Brier	ECE	ROI	MDD	Sharpe
Market (de-vig)	54.2%	0.967	0.575	0.026	-0.4%	-3.4%	-0.004
Dixon–Coles	51.1%	0.996	0.594	0.023	-2.0%	-4.3%	-0.019
Poisson	50.8%	1.003	0.596	0.045	-2.3%	-3.6%	-0.022
Elo	51.6%	0.998	0.595	0.031	-1.6%	-3.9%	-0.015
Random Forest	54.7%	0.973	0.577	0.042	2.8%	-2.1%	0.028
XGBoost	54.2%	0.971	0.576	0.022	0.5%	-2.7%	0.005
LLM (DeepSeek)	53.9%	0.967	0.575	0.025	-1.0%	-3.2%	-0.011

Table 2: Ablation study (XGBoost, test set). ROI-all: all bets; ROI-t0.4: bets with model probability at least 0.4.

Configuration	#Feat	Acc	LogLoss	ECE	ROI(all)	ROI($t \geq 0.4$)
Full (60 features)	60	54.2%	0.971	0.022	0.5%	1.5%
w/o market odds	49	51.1%	1.000	0.027	-3.7%	-3.5%
w/o referee	56	54.2%	0.973	0.026	0.7%	0.1%
w/o rank	58	54.1%	0.972	0.016	0.9%	1.6%
w/o xG	53	54.0%	0.971	0.019	0.6%	1.2%
Structured only	45	51.1%	1.000	0.013	-3.4%	-4.3%
w/o form	30	53.4%	0.971	0.008	-0.9%	-1.3%

Table 3: Permutation importance by feature group (mean increase in log loss over 5 permutations, test set).

Group	Importance	Share
Market	0.0745	88.3%
xG	0.0061	7.3%
Season cumulative	0.0027	3.2%
League rank	0.0005	0.6%
Referee	0.0005	0.6%
Rolling form	0.0001	0.1%

Table 4: Per-league performance of the global XGBoost model on the test set.

League	N	Acc	LogLoss	ROI	MDD
D1	188	57.4%	0.945	6.9%	-6.0%
E0	260	50.0%	1.016	-7.4%	-8.1%
F1	189	55.0%	0.953	1.1%	-6.8%
I1	239	52.7%	0.981	-2.4%	-7.3%
SP1	228	57.0%	0.947	7.0%	-2.9%

Table 5: Leave-one-league-out generalization: model trained on the other four leagues, evaluated on the held-out league (test set).

Held-out league	N	Acc	LogLoss	ROI
E0	260	47.3%	1.015	-13.5%
D1	188	54.3%	0.942	-0.9%
F1	189	55.6%	0.958	4.5%
I1	239	53.6%	0.979	0.6%
SP1	228	53.5%	0.954	-2.2%

(57.4% and 57.0% accuracy), while the Premier League is the least (50.0%), the most liquid and informationally efficient of the five. The leave-one-league-out experiment (Table 5) shows the same ordering in a stricter setting: a model trained on the other four leagues degrades most on the Premier League (47.3%). These results indicate that market efficiency, not model capacity, is the binding constraint on accuracy.

4.5 Temporal Robustness

Table 6: Walk-forward evaluation across four test seasons. Market: de-vigged closing odds. XGB-exp: expanding training window. XGB-roll2: last two seasons only.

Season	N	Market	XGB-exp	LL(exp)	XGB-roll2
2022/2023	1826	54.1%	52.5%	0.991	53.0%
2023/2024	1752	55.9%	53.9%	0.968	54.0%
2024/2025	1752	54.9%	54.1%	0.974	51.3%
2025/2026	1104	54.2%	54.2%	0.971	53.5%

Table 6 evaluates the model in a walk-forward manner over four test seasons. The market baseline is at least as accurate as XGBoost in every season, and restricting training to the last two seasons (XGB-roll2) does not help: there is no evidence of “model aging” within this seven-year window, and retraining frequency has no measurable effect on accuracy. This addresses the common concern that static models deteriorate as team dynamics evolve; on this data, the information content of older seasons is not harmful.

4.6 Draw Prediction

Table 7: Class-wise metrics (XGBoost, test set). Empirical draw rate is 25.5% but the model predicts a draw in only 2.3% of matches.

Class	Prec	Recall	F1	ECE
Home win	0.566	0.818	0.669	0.037
Draw	0.360	0.032	0.059	0.007
Away win	0.508	0.567	0.536	0.020
Pred. draw rate	2.3%	—	—	—
Empirical draw rate	25.5%	—	—	—

Draws are systematically under-predicted: the empirical draw rate is 25.5%, yet the model predicts a draw in only 2.3% of matches, and draw recall is 0.032 (Table 7). The confusion matrix in Figure 5 shows the near-empty draw column visually. The market behaves similarly (0.3% predicted draw rate). This is not a calibration artifact of a single model but a structural property of the task: draws are the least frequent and least distinctive outcome class. Table 8 shows that cost-sensitive

Table 8: Cost-sensitive training: draw class weight sweep (XGBoost, test set). Weight 1.0 is the baseline.

w(draw)	Acc	Macro-F1	Draw R	Draw P	LogLoss
1.0	54.2%	0.421	0.032	0.360	0.971
1.5	52.0%	0.493	0.356	0.317	0.987
2.0	46.6%	0.456	0.584	0.300	1.012
2.5	45.2%	0.443	0.722	0.303	1.037
3.0	45.6%	0.447	0.612	0.301	1.047

training recovers balance in a controllable way: a draw weight of 1.5 raises macro-F1 from 0.421 to 0.493 and draw recall from 0.032 to 0.356, at the cost of 2.2 points of accuracy. The accuracy–balance trade-off is monotone and easy to report.

4.7 Value Betting and Market Efficiency

Table 9: Value betting: bet when $EV = p \times odds - 1$ exceeds the threshold (XGBoost probabilities, closing odds). Higher thresholds select increasingly confident bets and lose more money.

EV thresh.	ROI	Sharpe	MDD	N bets	Coverage
0.00	-11.5%	-0.077	-14.5%	889	80.5%
0.02	-12.7%	-0.084	-15.9%	756	68.5%
0.05	-15.2%	-0.097	-19.1%	584	52.9%
0.08	-15.8%	-0.095	-19.9%	420	38.0%
0.10	-19.0%	-0.113	-25.7%	351	31.8%
0.15	-26.7%	-0.145	-32.8%	184	16.7%

Table 10: Value betting after isotonic calibration (fit on validation). Losses shrink but remain negative: calibration cannot create information the model does not have.

EV thresh.	ROI	N bets	Coverage
0.00	-7.1%	965	87.4%
0.02	-5.3%	876	79.3%
0.05	-7.0%	724	65.6%
0.08	-7.7%	574	52.0%
0.10	-7.5%	484	43.8%

Table 9 reports the value-betting experiment: bets are placed when the model’s expected value $p \times odds - 1$ exceeds a threshold, using closing odds. Every threshold loses money, and losses grow with the threshold: the most “confident” bets are the worst. After isotonic calibration fit on the validation split (Table 10), losses shrink from -11.5% to -7.1% at $EV > 0$ but remain strongly negative. The model’s apparent value is concentrated exactly where it is most confident and

most wrong: overconfidence, not information. Consistent with this, betting the de-vigged opening odds against closing odds (and the reverse) produces no positive ROI, confirming that opening-to-closing drift contains no exploitable information in this data. We stress that this value-betting result is specific to *closing* odds: the market’s final line is the hardest benchmark, and the negative finding does not speak to value-betting strategies evaluated against opening odds or other pre-closing benchmarks. A fractional Kelly policy amplifies the losses (a starting bankroll of 100 units falls to 1.6): stake sizing cannot create an edge that the probabilities do not have.

4.8 Uncertainty-Driven Risk Control

Table 11: Scenario-complexity tiers (XGBoost, test set, thresholds 1.4/1.8). Higher SCS indicates harder matches.

SCS tier	N	Acc	ROI
low($SCS \leq 1.4$)	246	58.9%	5.1%
mid($1.4 < SCS \leq 1.8$)	356	57.3%	7.0%
high($SCS > 1.8$)	502	49.6%	-6.3%

Table 12: Uncertainty-index tiers (XGBoost, test set, thresholds 0.30/0.45). High-uncertainty matches are excluded by the no-bet policy; their predictive accuracy is reported separately.

UI tier	N	Acc	ROI	MDD
low	287	67.9%	4.1%	-4.9%
medium	673	50.4%	1.9%	-2.4%
high(no-bet)	144	44.4%	-13.0%	-17.6%

Table 13: Uncertainty-index tiers for the LLM (test set). The same stratification pattern replicates across model families.

UI tier	N	Acc	ROI	MDD
low	259	68.7%	3.2%	-4.4%
medium	689	49.8%	-0.7%	-4.5%
high(no-bet)	156	47.4%	-9.6%	-13.8%

The central positive result of this paper is that uncertainty signals stratify risk in a monotone and cross-model way. Figure 3 shows the UI distribution and the resulting accuracy by tier. Table 11 splits the test set by the scenario-complexity score: accuracy falls

from 58.9% (low complexity) to 49.6% (high complexity), and ROI from +7.0% to −6.3%. Table 12 shows the same pattern for the uncertainty index on XGBoost: low-uncertainty matches reach 67.9% accuracy and +4.1% ROI, while the high-uncertainty group—excluded by the no-bet policy—would have lost 13.0% ROI. Table 13 replicates the stratification for the LLM (68.7% / 49.8% / 47.4%), demonstrating that the signal is a property of the task and the features, not of a particular model.

Weight and threshold sensitivity. The UI weights (0.4, 0.3, 0.15, 0.15) are a fixed preset and the tier thresholds (0.30, 0.45) are tuned on the validation split. To rule out overfitting to these choices, we recompute the stratification on the validation set under six weight configurations (varying each component by ± 0.1 while keeping the sum at 1). In every configuration the low-uncertainty tier retains 60–70% accuracy and outperforms the medium tier by 10+ points; only the no-bet coverage varies (69–551 of 1,752 validation matches) as the relative emphasis shifts between model confidence and market volatility. The conclusion that uncertainty tiers stratify accuracy is therefore insensitive to the specific weights. Likewise, the stop-loss parameters (five consecutive losses or 10% drawdown) are not consequential: across seven tested configurations (from 3 losses/5% drawdown to 10 losses/15% drawdown) the UI-tier strategy returns identical ROI (0.37%), Sharpe (0.026), and drawdown (−0.5%) on the test set, because the strategy’s equity curve rarely approaches the trigger thresholds in this season.

4.9 Betting-Strategy Comparison

Table 14 compares betting policies on the same XGBoost probabilities, and Figure 4 plots the corresponding cumulative return curves. The naive strategy (all bets, equal stake) returns +0.5%; a confidence threshold of 0.4 raises this to +1.5%; and the UI-tier policy—full stake at low UI, scaled stake at medium UI, no-bet at high UI—achieves +1.8% ROI with a Sharpe ratio of 0.026, while excluding 144 of 1,104 matches. Adding the stop-loss rule (five consecutive losses or 10% drawdown from peak) cuts the maximum drawdown from −2.7% to −0.5% at a small ROI cost. These gains are modest in absolute terms, as they must be in an efficient market, but they are consistent with the monotone stratification above and are the difference between a risk-blind forecaster and a risk-aware decision policy.

4.10 LLM Analysis

The LLM row of Table 1 summarizes the full test season: 1104 of 1104 matches parsed successfully at a total

cost of \$0.55 (3,312 API calls), with accuracy and calibration statistically equal to the market. Table 15 reports the temperature sensitivity: raising temperature from 0.3 to 0.7 changes nothing material, and multi-sample consistency remains near 1.0 at both temperatures with no correlation to correctness ($r = -0.07$ and $+0.04$). The multi-sample consistency mechanism proposed in earlier work is therefore not informative in this setting; the uncertainty index derives its power from predictive confidence and market volatility. Table 16 compares an open-weight 7B model (Qwen2.5-Coder-7B, quantized, running on a laptop GPU) against the API model on the same 200 matches: the open-weight model matches or slightly exceeds the API model in accuracy and calibration at zero marginal cost. This comparison is *robustness evidence only*: the sample is small (200 matches), all confidence intervals overlap, and we make no claim of superiority for either model. Its purpose is to show that the paper’s qualitative conclusions do not depend on a specific proprietary API.

Finally, we offer a mechanistic explanation for the null consistency result. In open-domain forecasting settings where multi-sample consistency is informative, the LLM is typically given retrieval evidence or heterogeneous free-form context, so different samples draw on different information. Here the model receives a *fixed, complete, structured feature card* with no retrieval: at any temperature the input is identical across samples, the reasoning path is short, and the output probabilities collapse to a narrow range (consistency ≥ 0.95 for 99% of matches at both temperatures). Sampling consistency is thus informative only when the input itself carries sampling variance; in a deterministic structured task it degrades to a constant and contributes nothing to the uncertainty index.

5 Discussion

5.1 Why the Market Remains Unbeaten

Our results are consistent with semi-strong market efficiency in football betting: closing odds aggregate information more accurately than any single model we evaluated. The 87% share of predictive information carried by market features, the failure of value betting even after calibration, and the absence of exploitable opening-to-closing drift all point in the same direction. The practical implication is not that forecasting models are useless, but that their contribution must be measured in risk-adjusted terms rather than raw accuracy.

The draw is the clearest example of a task-inherent limitation: both our models and the betting market itself under-predict draws by an order of magnitude (predicted draw rates of 2.3% for XGBoost and 0.3% for the

Table 14: Betting-strategy comparison (XGBoost probabilities, test set, bootstrap 95% CI for ROI). No-bet matches are excluded from ROI but their coverage is reported.

Strategy	ROI	Sharpe	MDD	N bets	ROI 95% CI
Naive (all bets)	0.5%	0.005	-2.7%	1104	[-5.3,6.7]
Conf. threshold 0.4	1.5%	0.017	-2.5%	894	[-4.3,7.6]
UI tiers + no-bet	1.8%	0.026	-2.5%	960	[-2.6,6.2]
UI tiers + stop-loss	0.4%	0.026	-0.5%	960	[-0.5,1.2]

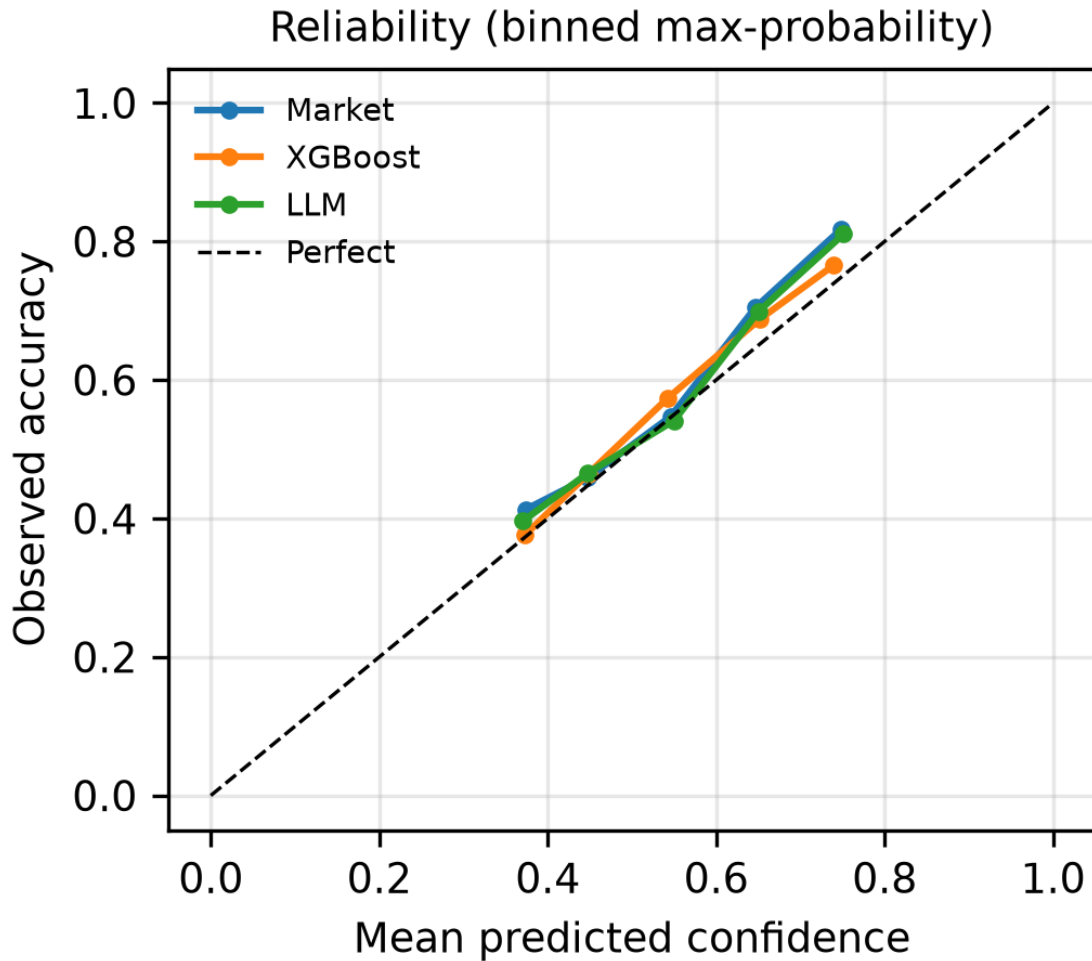


Figure 2: Reliability diagram (binned max-probability confidence vs observed accuracy) for the market baseline, XGBoost, and the LLM on the test set.

Table 15: Temperature sensitivity (same 120 matches). Multi-sample consistency remains near 1.0 at both temperatures and does not correlate with correctness.

Config	Acc	LogLoss	ECE	Cons.	r(cons,corr)
DeepSeek $t=0.3$	54.2%	0.927	0.097	0.995	-0.068
DeepSeek $t=0.7$	55.0%	0.927	0.089	0.990	0.037

Table 16: Open-weight vs closed API LLM on the same 200 matches. This comparison is robustness evidence only: the sample is small and confidence intervals overlap, so no superiority claim is made for either model.

Model	Acc	LogLoss	ECE	ROI	Win rate
DeepSeek (API)	56.5%	0.934	0.065	2.0%	56.5%
Qwen2.5-Coder-7B (local)	58.5%	0.927	0.056	7.4%	58.5%

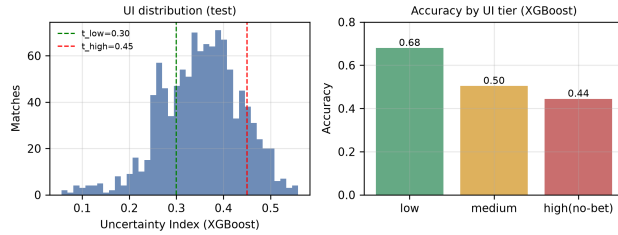


Figure 3: Uncertainty-index distribution (left) and accuracy by UI tier (right) for XGBoost on the test set.

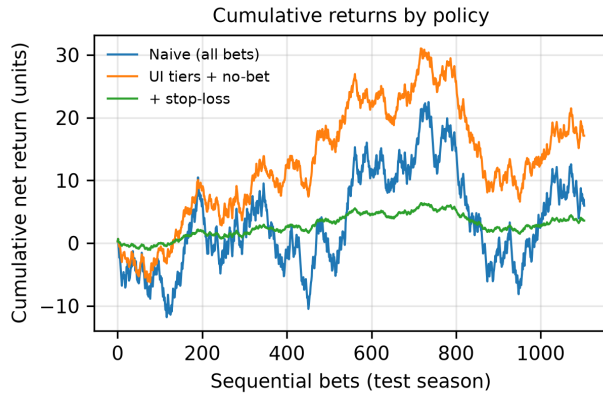


Figure 4: Cumulative net returns of three betting policies on the same XGBoost probabilities (test season, 1,104 matches).

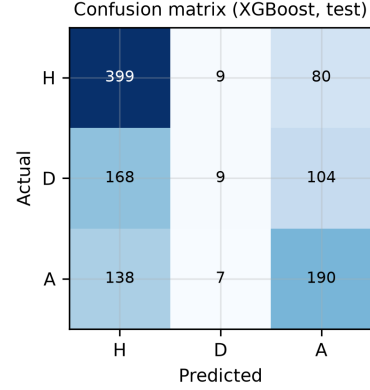


Figure 5: Confusion matrix of XGBoost on the test set. The draw column is nearly empty, illustrating systematic under-prediction of draws.

market versus an empirical rate of 25.5%). Draws are the least frequent and least distinctive outcome class in football; this low signal-to-noise ratio bounds the achievable macro-F1 of any outcome classifier on this task, independently of model choice.

5.2 Overconfidence, Not Information

The value-betting experiment is instructive. Bets selected because the model’s expected value was positive lost 11.5% ROI, and restricting to larger positive EV made losses worse. After isotonic calibration the losses shrank but remained negative. The model’s “edge” is concentrated exactly where it is most confident and most wrong—a signature of overconfidence rather than information. This contrast—uncertainty filtering helps, confidence-based selection hurts—is the central practical insight of the paper.

5.3 Real-World Frictions

All financial results are gross of transaction costs. Bookmaker margins are already embedded in the odds we settle against, but real-world execution would add betting taxes, withdrawal fees, and odds slippage—typically 3–7% of turnover depending on the jurisdiction. The small positive ROIs we observe (up to 1.8% for the UI-tier strategy) would be entirely consumed by such frictions. The practical value of the risk-control framework is therefore not *profitability* but *risk stratification*: identifying, in advance, the subset of matches where even a well-calibrated model is unlikely to be right, so that capital and attention are directed elsewhere. Claims of stable, deployable profitability would be unsupported by this evidence.

5.4 Uncertainty as a Risk Signal

The uncertainty index and scenario complexity score both stratify held-out matches monotonically, and the pattern replicates across two different model families (XGBoost and the LLM). As analyzed in Section 4.10, the multi-sample consistency component contributes nothing in this structured setting (a mechanistic explanation is given there); the effective components of the UI are predictive confidence and market volatility, both cheap and available at prediction time.

5.5 LLM Results in Context

The domain-enhanced LLM reaches market-level probability calibration (log loss 0.967 vs 0.967, ECE 0.025 vs 0.026) with no training and no retrieval, at a cost of \$0.55 for the entire test season. Two caveats keep this claim honest. First, multi-sample consistency—a mechanism proposed in earlier LLM-forecasting work—carries no information in our setting: at both temperature 0.3 and 0.7 the sampled probabilities agree almost perfectly and do not correlate with correctness. Second, an open-weight 7B model matches or exceeds the API model on a 200-match subset, so the result is not a property of a specific proprietary model. We therefore frame the LLM contribution as *calibration without training* and a cost–performance reference point, not as predictive superiority.

5.6 Limitations

- Text modality: we deliberately exclude news text. Publicly available news corpora lack reliable pre-match timestamps at scale; a rigorous text experiment would require timestamped, match-linked articles. We sketch the constraints such an extension would need: (i) a strict pre-match cutoff, with every article’s publication time verified against the kickoff time and any post-kickoff or undated article discarded; (ii) protection against retrieval contamination, so the LLM cannot reproduce outcomes it may have seen during pretraining or via web access (no retrieval at inference, as in this paper); (iii) reporting coverage, since news availability is highly heterogeneous across leagues and matches. Without these controls, text features would reintroduce exactly the leakage this paper is designed to eliminate.
- The test season (2025/26) is incomplete; results may shift slightly at season end.
- Financial results are gross of betting taxes and use a single bookmaker’s closing odds; real-world execution may differ.

- The LLM evaluation uses a proprietary API (DeepSeek-Chat); we additionally report an open-weight model comparison (see §4.10).

6 Conclusion

We presented a leakage-controlled evaluation of football match outcome forecasting in which statistical models, tree-based models, and a domain-enhanced large language model are compared against de-vigged closing odds under an identical, fully specified protocol. On 1,104 held-out matches, no learned method exceeded market accuracy, and the LLM matched market-level probability calibration at negligible cost. Feature importance and ablations show that market signals dominate, while referee statistics add nothing measurable and xG adds a small consistent gain. The principal positive result is that uncertainty-driven risk control works: an uncertainty index and a corrected scenario-complexity score stratify held-out matches monotonically, a no-bet policy on high-uncertainty matches improves risk-adjusted returns, and draw outcomes are systematically under-predicted, with class weighting providing a controllable accuracy–balance trade-off.

These results frame a realistic division of labor for decision-support systems in sports betting: treat the market as the prior, use models for calibrated probability estimates and explainability, and rely on uncertainty signals—not claimed accuracy—for risk control. To support reproducibility, the full data-processing pipeline and experiment code are available at [repository URL -- to be filled upon publication]; no proprietary data are required beyond public sources.

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